UNIT-1

UNIT - I:

Information Theory: Shanon Capacity, Multimedia Data, Data Processing, Boolean Logics, Information Content, Entropy, Source Coding, Channel Coding, Modulation Schemes, Internet.

1) Shannon Capacity

- **Definition:** The maximum data rate at which information can be transmitted over a communication channel with a very low probability of error.
- Proposed by Claude Shannon.
- Based on **Shannon–Hartley Theorem**.
- **Formula:**

  C=Blog2(1+S/N)C = B \log_2(1 + S/N)C=Blog2(1+S/N)

  Where:

  - C = Channel Capacity (bits/sec)
  - B = Bandwidth (Hz)
  - S/N = Signal-to-Noise Ratio
- **Key Points:**
  - Higher bandwidth → higher capacity
  - Higher SNR → higher capacity
  - Capacity is theoretical maximum
- **Importance:** Foundation of modern digital communication systems.

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2) Multimedia Data

- **Definition:** Data that includes a combination of text, audio, images, video, and animation.
- **Types:**

    - Text
    - Audio
    - Images
    - Video
    - Graphics
- **Characteristics:**
    - Large file size
    - Requires compression
    - Time-dependent (audio/video)
- **Applications:**
    - Video conferencing
    - Online streaming
    - E-learning
- **Challenges:**
    - High bandwidth requirement
    - Storage and processing power

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3) Data Processing

- **Definition:** The collection and manipulation of data to produce meaningful information.
- **Steps:**
    1. Data collection
    2. Data input
    3. Processing
    4. Output
    5. Storage
- **Types:**
    - Batch processing
    - Real-time processing
    - Online processing
- **Applications:**
    - Banking systems
    - Payroll systems
    - Inventory management

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4) Boolean Logic

- **Definition:** A form of algebra in which all values are either TRUE (1) or FALSE (0).
- Developed by George Boole.
- **Basic Operations:**
    - AND

    - OR
    - NOT
- **Other Operations:**
    - NAND
    - NOR
    - XOR
- **Applications:**
    - Digital circuits
    - Computer programming
    - Decision making systems

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5) Information Content

- **Definition:** The amount of information produced by a particular event.
- Measured in **bits**.
- Formula:

  I(x)=−log2P(x)I(x) = -\log_2 P(x)I(x)=−log2P(x)

  Where P(x) = Probability of event x.

- **Key Idea:**
    - Rare events → more information
    - Common events → less information
- **Example:**
    - Getting “Head” in fair coin toss → 1 bit

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6) Entropy

- **Definition:** The average information content of a message source.
- Introduced by Claude Shannon.
- **Formula:**

  H(X)=−∑P(x)log2P(x)H(X) = - \sum P(x) \log_2 P(x)H(X)=−∑P(x)log2P(x)

- **Unit:** Bits per symbol.
- **Key Points:**
    - Measures uncertainty
    - Maximum when all outcomes are equally likely
    - Minimum when outcome is certain

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7) Source Coding

- **Definition:** The process of encoding information efficiently to reduce redundancy.
- Also called **data compression**.
- **Types:**
  - Lossless coding (e.g., Huffman coding)
  - Lossy coding (e.g., JPEG, MP3)
- **Purpose:**
  - Reduce storage
  - Reduce transmission bandwidth
- **Example Techniques:**
  - Run-length encoding
  - Huffman coding

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8) Channel Coding

- **Definition:** Adding extra bits to data to detect and correct errors during transmission.
- Also called **error control coding**.
- **Types:**
  - Error detection (Parity bit, CRC)
  - Error correction (Hamming code, Reed-Solomon)
- **Purpose:**
  - Improve reliability
  - Reduce transmission errors
- Used in wireless communication, satellite, and mobile networks.

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9) Modulation Schemes

- **Definition:** Technique of modifying a carrier signal to transmit data.
- **Types of Modulation:**
  - Amplitude Modulation (AM)
  - Frequency Modulation (FM)
  - Phase Modulation (PM)
- **Digital Modulation:**
  - ASK (Amplitude Shift Keying)
  - FSK (Frequency Shift Keying)
  - PSK (Phase Shift Keying)
  - QAM (Quadrature Amplitude Modulation)
- **Purpose:**

    - Efficient transmission
    - Long-distance communication
- Used in radio, TV, Wi-Fi, and mobile networks.

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10) Internet

- **Definition:** A global network of interconnected computer networks.
- Based on **TCP/IP protocol suite**.
- Developed from ARPANET.
- **Services:**
    - Email
    - Web browsing
    - File transfer
    - Video streaming
- **Components:**
    - ISPs
    - Routers
    - Servers
- **Advantages:**
    - Global communication
    - Information access
- **Limitations:**
    - Security threats
    - Privacy concerns

UNIT-2

UNIT - II:

Wireless Communication Technologies: WLAN, Wifi, Bluetooth, Other Wireless PAN And WAN Technologies, Satellite Communications, Broadcast Services.

1) WLAN (Wireless Local Area Network)

- **Definition:** A wireless network that connects devices within a limited area such as homes, offices, schools, or campuses.
- **Standard:** Based on IEEE 802.11.
- **Coverage Range:** Typically 30–100 meters indoors.
- **Frequency Bands:** 2.4 GHz, 5 GHz, and 6 GHz (newer standards).
- **Components:**
  - Access Point (AP)
  - Wireless Network Interface Card (WNIC)
  - Router
- **Advantages:**
  - Easy installation
  - Mobility within coverage area
  - Cost-effective
- **Limitations:**
  - Limited range
  - Interference from other wireless devices
  - Security risks if not properly configured

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2) Wi-Fi

- **Definition:** A wireless networking technology that allows devices to connect to the internet or communicate wirelessly within a WLAN.
- **Based on:** Wi-Fi (built on IEEE 802.11 standards).
- **Common Standards:**
  - 802.11n (Wi-Fi 4)
  - 802.11ac (Wi-Fi 5)
  - 802.11ax (Wi-Fi 6)
- **Speed:** Ranges from Mbps to several Gbps depending on the standard.
- **Security Protocols:**
  - WPA2
  - WPA3
- **Applications:**
  - Internet access
  - File sharing
  - Video streaming
  - Smart home devices

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3) Bluetooth

- **Definition:** A short-range wireless communication technology used for device-to-device communication.
- **Standard:** Based on Bluetooth.
- **Range:**
  - Class 2: ~10 meters
  - Class 1: ~100 meters
- **Frequency Band:** 2.4 GHz ISM band.
- **Data Rate:** Up to 2 Mbps (Classic), higher with BLE versions.
- **Types:**
  - Bluetooth Classic
  - Bluetooth Low Energy (BLE)
- **Applications:**
  - Wireless headphones
  - Keyboards and mice
  - File transfer
  - IoT devices
- **Advantages:**
  - Low power consumption
  - Easy pairing
- **Limitations:**
  - Short range
  - Lower speed compared to Wi-Fi

4) Other Wireless PAN and WAN Technologies

(A) Wireless PAN (Personal Area Network)

- Covers very small area (within 10 meters).
- Technologies:
  - Bluetooth
  - Zigbee
  - Infrared (IR)
- Used in wearables, smart devices, and sensor networks.

(B) Wireless WAN (Wide Area Network)

- Covers large geographical areas (cities, countries).
- Technologies:
  - 3G
  - 4G LTE
  - 5G
- Provides mobile internet access.
- Operated by telecom service providers.
- Advantages: Wide coverage and mobility.
- Limitations: Higher cost, network congestion.

5) Satellite Communications

- **Definition:** Communication using artificial satellites to relay signals.
- **Orbit Types:**
  - GEO (Geostationary Earth Orbit)
  - MEO (Medium Earth Orbit)
  - LEO (Low Earth Orbit)
- **Components:**
  - Ground station
  - Satellite transponder
  - User terminal
- **Applications:**
  - TV broadcasting
  - GPS
  - Weather forecasting
  - Remote area internet
- **Advantages:**
  - Global coverage
  - Useful in remote areas
- **Limitations:**
  - High latency (especially GEO)
  - Expensive infrastructure

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6) Broadcast Services

- **Definition:** One-to-many communication system where signals are transmitted to a large audience.
- **Types:**
  - Radio broadcasting
  - Television broadcasting
- **Transmission Medium:**
  - Terrestrial transmitters
  - Satellite
  - Cable networks
- **Examples:**
  - FM/AM radio
  - Digital TV (DTH)
- **Advantages:**
  - Wide audience reach
  - Efficient mass communication
- **Limitations:**
  - One-way communication
  - Limited personalization

UNIT-3

UNIT - III:

Cellular Mobile Networks: GSM(2G), UMTS (3G), LTE(4G), 5G Mobile Networks, Mobile Network Planning Aspects.

1) GSM (2G)

- **Full Form:** Global System for Mobile Communications
- **Generation:** Second Generation (2G) mobile network
- Developed by European Telecommunications Standards Institute (ETSI).
- **Technology Type:** Digital cellular network
- **Frequency Bands:** 900 MHz, 1800 MHz (commonly)
- **Multiple Access Technique:**
  - TDMA (Time Division Multiple Access)
  - FDMA (Frequency Division Multiple Access)
- **Services Provided:**
  - Voice calls
  - SMS (Short Message Service)
  - Limited data (GPRS, EDGE)
- **Data Speed:** Up to 384 kbps (with EDGE)
- **Advantages:**
  - Better voice quality than 1G
  - International roaming
  - SIM-based authentication
- **Limitations:**
  - Low data speed
  - Limited multimedia support

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2) UMTS (3G)

- **Full Form:** Universal Mobile Telecommunications System
- **Generation:** Third Generation (3G)
- Standardized by 3rd Generation Partnership Project (3GPP).

- **Technology Type:** WCDMA (Wideband Code Division Multiple Access)
- **Data Speed:**
  - Up to 2 Mbps (initially)
  - Higher speeds with HSPA/HSPA+
- **Services:**
  - Video calling
  - Mobile internet
  - Multimedia messaging
- **Improvements over 2G:**
  - Higher data rates
  - Better spectrum efficiency
- **Advantages:**
  - Supports multimedia services
  - Global roaming
- **Limitations:**
  - Higher infrastructure cost
  - Higher power consumption

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3) LTE (4G)

- **Full Form:** Long Term Evolution
- **Generation:** Fourth Generation (4G)
- Developed by 3rd Generation Partnership Project (3GPP).
- **Technology Type:**
  - OFDMA (Downlink)
  - SC-FDMA (Uplink)
- **Data Speed:**
  - Up to 100 Mbps (mobile)
  - Up to 1 Gbps (theoretical)
- **Features:**
  - All-IP network architecture
  - Low latency
  - High spectral efficiency
- **Services:**
  - HD video streaming
  - VoLTE (Voice over LTE)
  - Online gaming
- **Advantages:**
  - High speed
  - Low delay
  - Better capacity
- **Limitations:**
  - Requires new infrastructure
  - Higher deployment cost

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4) 5G Mobile Networks

- **Generation:** Fifth Generation mobile network
- Standardized by 3rd Generation Partnership Project (3GPP).
- **Key Features:**
  - Ultra-high speed (up to 10 Gbps theoretical)
  - Ultra-low latency (~1 ms)
  - Massive device connectivity
- **Technology Components:**
  - Massive MIMO
  - Beamforming
  - Millimeter wave (mmWave)
- **Use Cases:**
  - Smart cities
  - Autonomous vehicles
  - IoT (Internet of Things)
  - Remote surgery
- **Advantages:**
  - Very high data rate
  - Reliable communication
  - Supports large number of devices
- **Challenges:**
  - High infrastructure cost
  - Limited coverage of mmWave
  - Requires dense small cell deployment

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5) Mobile Network Planning Aspects

- **Definition:** The process of designing and optimizing cellular networks for efficient coverage and capacity.
- **Key Planning Factors:**

1. **Coverage Planning:**
    - Ensuring signal availability in all required areas
2. **Capacity Planning:**
    - Supporting expected number of users
3. **Frequency Planning:**
    - Efficient spectrum utilization
    - Avoiding interference
4. **Cell Planning:**
    - Cell size (Macro, Micro, Pico, Femto cells)
5. **Handover Planning:**
    - Seamless transition between cells
6. **Quality of Service (QoS):**
    - Maintaining call quality and data speed

- **Important Considerations:**
    - Terrain and building structures
    - Population density
    - Traffic demand
    - Regulatory constraints
- **Goal:**
    - Maximize coverage
    - Minimize interference
    - Optimize cost and performance